



Effects of mussel-kelp ratios in integrated mariculture on the carbon dioxide system in Sanggou Bay

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ABSTRACT

The ocean is the largest carbon reservoir on Earth. The shellfish-seaweed mode of Integrated Multi-trophic Aquaculture (IMTA) can be optimized to enhance carbon removal from the sea. In this study, the effects of different mussel-kelp ratios (i.e. 6:3, 6:2, 6:1, 4:3, 4:2, 4:1, 2:3, 2:2, and 2:1) on water quality and the carbon dioxide (CO₂) system were investigated using experimental enclosures in Sanggou Bay, China. The results showed that the DO decreased significantly when ratios were 4:1, 6:2 and 6:1. With the exception of ratio 4:3, there was no significant change in pH in any treatment. The highest of DO, pH, and CO₃²⁻ values occurred when the mussel-kelp ratio was 2:3, the highest proportion of kelp, while the lowest occurred in ratio 6:1, with the highest proportion of mussels. The highest levels of pCO₂, CO₂ and HCO₃⁻ were observed in ratio 6:1, and the lowest when the ratio was 2:3. Both DIC and HCO₃⁻ significantly decreased by the end of the experiment for ratios 2:1 and 6:3, while the concentration of CO₂ significantly decreased for ratios 2:3 and 6:3. Considering the measured parameters, the mussel-kelp ratios of 2:1, 2:2, 4:2 and 4:3 were identified as the best net carbon sinks with the smallest negative environmental impacts. This study revealed how the IMTA shellfish-seaweed modes can be optimized to mitigate ocean acidification using specific proportions of species.

1. Introduction

The anthropogenic release of CO₂ into the atmosphere has had negative consequences for environment. If current emission trends continue unabated, it has been estimated that global temperatures will increase by about 1.5 °C between 2030 and 2052, causing longer heat waves and more frequent extreme precipitation (Petrović and Lobanov, 2020). The ocean, which possesses a complex carbon cycle, is a major sink for anthropogenic CO₂ and had absorbed roughly 30% of anthropogenic CO₂ emissions from the beginning of the industrial revolution to the mid-1990s (Gruber et al., 2019; Takahashi et al., 2002). In 2018, total anthropogenic CO₂ emissions were estimated to be 11.5 ± 0.9 GtC, of which 2.6 ± 0.6 GtC was absorbed by the ocean (Friedlingstein et al., 2019). The biological carbon pump of the ocean is an important process that transports approximately three-quarters of the dissolved inorganic carbon (DIC) from the surface to the deep-ocean (Riebesell et al., 2007). The areas with the largest proportional contribution to the oceanic

carbon cycle are the highly complex marginal seas, affected by human activities, complex currents, terrestrial nutrients sources, etc. (Borges, 2005). The biological carbon cycle is especially active in these regions, with the natural planktonic community, bacteria and invertebrate playing larger roles in the carbon cycle in the marginal seas (Hong et al., 1997; Huang et al., 1999; Xiao and Wang, 2000; Tang et al., 2013). Fisheries and aquaculture activities can improve the carbon absorption capacity in marginal seas by removing carbon from the coastal ecosystem via harvesting (Tang et al., 2011). In 2018, global fisheries and aquaculture production reached 178.5 million tons, of which 64.54% originated from marine areas. Total aquaculture production alone reached 82.1 million tons, 78.66% of which came from China (FAO, 2020).

However, aquaculture can have a variety of negative environmental impacts (Gillibrand et al., 1996; Wen et al., 2007). To overcome these challenges, integrated multi-trophic aquaculture (IMTA) approaches have been implemented worldwide to sustainably develop aquaculture

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(Chopin et al., 2010). Several IMTA modes have been utilized with great success, including fish-seaweed (Abreu et al., 2009; Fossberg et al., 2018), finfish-shellfish (Langan, 2004; Cubillo et al., 2016), abalone-kelp, finfish-bivalve-kelp, abalone-sea cucumber-clam-seaweed (Fang et al., 2016), and shellfish-seaweed (Fang et al., 2019). Based on their nutrient requirements and waste products, species for co-culture in IMTA can be divided into different niches. By combining species with complementary functions, aquaculture systems can be made healthy and sustainable (Reid et al., 2020). For example, seaweeds in IMTA can absorb soluble nutrients, such as ammonium (NH_4^+), nitrate (NO_3^-), phosphate (PO_4^{3-}) and carbon dioxide (CO_2) (Reid et al., 2020), therefore mitigating the eutrophication of the aquaculture system. Reid et al. (2013) found that the amount of carbon respired by 1 kg of Atlantic salmon (*Salmo salar*) could be absorbed by 10.2 kg of kelp (*Saccharina latissima*) in the fish-seaweed IMTA mode. In the shellfish-seaweed mode no additional feed is required, making it an efficient and economically important mode that benefits the environment and carbon cycle (Tang et al., 2011). Seaweed absorbs DIC through photosynthesis, and suspension-feeding shellfish consume particulate organic carbon (POC). At the same time, shellfish integrate a large amount of carbon into their shells in the form of CaCO_3 during calcification. Han et al. (2013) found that the different proportions of scallop (*Chlamys farreri*) and seaweed (*Gracilaria lemaneiformis*) in the shellfish-seaweed mode demonstrated varying effects on the CO_2 system in sea water. However, studies like this, related to the collocation ratios of shellfish to seaweed in the shellfish-seaweed mode, are limited.

Sanggou Bay ($37^\circ 01' - 37^\circ 09' \text{ N}$, $122^\circ 24' - 122^\circ 35' \text{ E}$), located in the Shandong Province of China, is a well-known aquaculture area for IMTA due to its suitable environmental conditions. There are over 30 important aquaculture species in this bay, with shellfish and seaweed being the most commonly cultured groups (Zhang et al., 2007). It has been shown that shellfish-seaweed combination, such as the mode of mussel-kelp, is a highly successfully IMTA mode, and also is the base of such IMTA modes (Tang et al., 2013; Fang et al., 2016). However, the proportions of shellfish to seaweed can have various effects on sea water parameters, and using a suitable appropriate proportion of the two species is important for a balanced aquaculture environment (Han et al., 2013). Therefore, in this study, the effects of different mussel-kelp ratios on the CO_2 system in sea water were examined with the aim of determining the optimal starting proportion of mussels to kelp when initiating a mussel-kelp culture. This effort will help establish of a sustainable aquaculture practices in Sanggou Bay. The results can serve as a reference for aquaculture operations in different geographic areas.

2. Material and methods

2.1. Experimental design and management

The experiment was conducted in Sanggou Bay, China. The kelp seeding *Laminaria japonica* and mussels *Crassostrea gigas* were obtained from Sanggou Bay. The kelps ($2.61 \pm 1.34 \text{ g}$) were tied to ropes, and the mussels ($14.20 \pm 0.97 \text{ g}$) were cleaned and placed in a netted bag. Both of the organisms were hung in a transparent polyethylene plastic bag (1.0 m^3) that was closed with a cap without air bubbles. Each bag was connected to a special cap that was easy to open and close. According to different collocations of mussel and kelp (Table 1), nine different ratios were established. They were recorded as 6:3, 6:2, 6:1, 4:3, 4:2, 4:1, 2:3, 2:2, 2:1, according to the body weight of mussel (300 g, 200 g, 100 g) and kelp (150 g, 100 g, 50 g). Three replications were used in each ratio.

Bags containing mussel and kelp were tied to a floating frame ($10 \text{ m} \times 10 \text{ m}$) in the sea, and were filled with natural seawater. Samples were collected at 17:00 on the first day, and 6:00 and 17:00 on both the second and third days. Water temperature, salinity, pH, and dissolved oxygen (DO) were measured by YSI Professional Plus (Pro Plus, USA) and pH meter (Orion 9107 BNMD, USA) at the same time. The duration of each sample time was approximately 1.5 h. The water samples were

Table 1

Different ratios of mussels integrated with kelp.

Ratios	Mussel (g)	Kelp (g)
6:3	300	150
6:2	300	100
6:1	300	50
4:3	200	150
4:2	200	100
4:1	200	50
2:3	100	150
2:2	100	100
2:1	100	50

filtered using cellulose acetate filters ($0.45 \mu\text{m}$, Whatman). After that, 150 ml of filtered sea water was mixed with 0.02% volume of HgCl_2 saturated solution for total alkalinity (TA) analysis, and stored in sealed 150-mL borosilicate bottles at 4°C (Dickson and Goyet, 1994).

2.2. Samples analysis

Total alkalinity (TA) was measured using an automatic titrator (848 Titrino plus, Switzerland). Parameters for the seawater carbonate system were calculated from pH, total alkalinity, temperature, and salinity according to the method of Dickson and Goyet (1994).

$$\text{CA} = \text{TA} - \text{BT} \times \frac{K' B}{a\text{H}^+ + K' B}$$

where CA and B_T represent the carbonate alkalinity and concentration of boric acid in seawater, respectively (Millero et al., 1998). $a\text{H}^+$ is approximated as hydrogen ion concentration in seawater. $K' B$ is the dissociation constants for boric acid (Dickson, 1990).

$$\text{DIC} = \text{CA} \times \frac{a\text{H}^+ + K'_2 + \frac{(a\text{H}^+)^2}{K'_1}}{a\text{H}^+ + 2 \times K'_2}$$

$$[\text{HCO}_3^-] = \text{CA} \times \frac{a\text{H}^+}{a\text{H}^+ + 2 \times K'_2}$$

$$[\text{CO}_3^{2-}] = \text{CA} \times \frac{K'_2}{a\text{H}^+ + 2 \times K'_2}$$

$$[\text{CO}_{2(\text{T})}] = \text{CA} \times \frac{(a\text{H}^+)^2}{K'_1 \times (a\text{H}^+ + 2 \times K'_2)}$$

$$\text{pCO}_2 = \frac{\text{CO}_{2(\text{T})}}{\alpha}$$

where K'_1 and K'_2 represent the temperature and salinity dependent the first and second dissociation constants for carbonic acid, respectively (Mehrbach et al., 1973). α is the CO_2 solubility constant (Weiss, 1974).

The mussel and kelp were removed from the bags and wrapped with sterile gauze to remove external water, and then dried separately in an oven at 70°C for 48 h to measure the dry weight after the field experiment.

2.3. Data analysis

Statistical analysis of the data was performed using SPSS 13.0 for Windows (IBM Corp., NY, USA). The normality of distribution and homogeneity of variances were tested for the data. A log-transformation was applied when necessary. The mussel and kelp in each ratio were considered together as a combined system. The influence of, and interaction, between biomass ratios and sampling time on the DO, pH and parameters of the carbonate system were tested using two-way analyses of variance (ANOVA). If an interaction between the two main factors in the two-way ANOVA model was significant, one-way ANOVA was used

to examine the effects of each of the main factors on the physiological parameters. A Bonferroni adjustment was applied to all parameters to avoid the magnification of statistical errors resulting from the repeated use of a one-way ANOVA. The differences among the initial body weight, dry weight of organisms were tested using a one-way ANOVA. The value of $P < 0.05$ was considered to be statistically significant.

3. Results

During the experimental period, the water temperature was 8.9–10.3 °C, and salinity was 31.43–31.59. No mortality was observed during the experiment. To track the biomass of the organisms, we measured the dry weight of the mussel soft tissue and kelp as described in the experimental design (Table 2).

The two-way ANOVA analysis of the parameters from the different mussel-kelp ratio treatments at different sampling times showed simple and regular patterns (Table 3 and Table 4). The DO, pH, $p\text{CO}_2$ and CO_2 system were significantly affected by both mussel-kelp ratio and time ($P < 0.05$). With the exception of pH, the interactions between mussel-kelp ratio and time were significant for all parameters ($P < 0.05$). Levels of DO, pH, and CO_3^{2-} were the highest when the mussel-kelp ratio was 2:3, which represented the highest proportion of kelp ($P < 0.05$), and lowest when the ratio was 6:1, representing the highest proportion of mussels ($P < 0.05$). In contrast, $p\text{CO}_2$, CO_2 , and HCO_3^- levels were highest when the mussel-kelp ratio was 6:1 and lowest when the ratio was 2:3 ($P < 0.05$). At the end of the experiment, DO, $p\text{CO}_2$, and CO_2 had decreased significantly ($P < 0.05$), while increases in pH, CO_3^{2-} , HCO_3^- , and DIC were observed. However, there was no significant difference in pH between the beginning and the end of the experiment ($P > 0.05$).

3.1. Dissolved oxygen

The dissolved oxygen (DO) exhibited significant diurnal fluctuation, decreasing during the night and increasing during the day (Fig. 1). At the end of the experiment, the DO had significantly decreased in treatments with mussel-kelp ratios of 4:1, 6:2 and 6:1, but had not changed in all other ratio treatments (one-way ANOVA: 4:1, $F_{4,10} = 13.27$, $P < 0.05$; 6:2, $F_{4,10} = 16.65$, $P < 0.05$; 6:1, $F_{4,10} = 19.64$, $P < 0.05$).

3.2. pH

As shown in Fig. 2, the pH did not change much during the experiment and, with the exception of the mussel-kelp ratio of 4:3, there was

Table 2

The biomass of organisms in the experiment (means $\pm$ S.E.)

Ratios	Mussel No. (ind)	DW _s (g)	BW _{wet} (g)	Kelp weight (g)	Kelp dry weight (g)
6:3	20.33 $\pm$ 0.33 ^c	17.62 $\pm$ 0.52 ^c	300.17 $\pm$ 0.59 ^c	150.19 $\pm$ 1.11 ^c	10.74 $\pm$ 0.08 ^c
6:2	20.00 $\pm$ 0.58 ^c	17.41 $\pm$ 0.33 ^c	300.34 $\pm$ 0.58 ^c	99.57 $\pm$ 0.32 ^b	7.13 $\pm$ 0.02 ^b
6:1	19.67 $\pm$ 0.33 ^c	17.61 $\pm$ 0.26 ^c	300.09 $\pm$ 0.71 ^c	49.80 $\pm$ 0.76 ^a	3.16 $\pm$ 0.05 ^a
4:3	15.00 $\pm$ 0.58 ^b	13.15 $\pm$ 0.64 ^b	200.95 $\pm$ 1.23 ^b	149.17 $\pm$ 0.57 ^c	10.61 $\pm$ 0.04 ^c
4:2	15.33 $\pm$ 0.33 ^b	13.56 $\pm$ 0.45 ^b	200.37 $\pm$ 0.66 ^b	99.28 $\pm$ 0.55 ^b	7.10 $\pm$ 0.04 ^b
4:1	15.00 $\pm$ 0.58 ^b	13.33 $\pm$ 0.49 ^b	200.26 $\pm$ 1.03 ^b	50.37 $\pm$ 0.46 ^a	3.54 $\pm$ 0.03 ^a
2:3	7.33 $\pm$ 0.33 ^a	5.81 $\pm$ 0.40 ^a	100.64 $\pm$ 1.00 ^a	149.75 $\pm$ 0.57 ^c	10.65 $\pm$ 0.04 ^c
2:2	7.00 $\pm$ 0.00 ^a	5.94 $\pm$ 0.31 ^a	100.45 $\pm$ 0.75 ^a	99.28 $\pm$ 0.55 ^b	7.01 $\pm$ 0.04 ^b
2:1	7.33 $\pm$ 0.33 ^a	5.85 $\pm$ 0.38 ^a	100.68 $\pm$ 0.97 ^a	50.67 $\pm$ 0.53 ^a	3.62 $\pm$ 0.04 ^a

Notes: DW_s: Dry weight of mussel soft part; BW_{wet}: Wet body weight of mussel.

Table 3

Two-way ANOVA results of dissolved oxygen (DO), pH, $p\text{CO}_2$ of mussel-kelp modes at different sampling time.

Source		df	Mean Squares	F	P
DO	Model	44	1.39	11.55	< 0.001
	Ratio	8	3.03	25.24	< 0.001
	Time	4	5.98	49.81	< 0.001
	Ratio* Time	32	0.40	3.34	< 0.001
	Error	90	0.12		
	Total	135			
pH	Model	44	0.01	2.22	0.001
	Ratio	8	0.03	5.15	< 0.001
	Time	4	0.02	3.59	0.009
	Ratio* Time	32	0.01	1.31	0.160
	Error	90	0.01		
	Total	135			
$p\text{CO}_2$	Model	44	16,583.01	22.33	< 0.001
	Ratio	8	37,138.78	50.00	< 0.001
	Time	4	30,679.23	41.31	< 0.001
	Ratio* Time	32	9682.05	13.04	< 0.001
	Error	90	742.74		
	Total	135			

Table 4

Two-way ANOVA results of CO_2 , CO_3^{2-} , HCO_3^- , dissolved inorganic carbon (DIC) in the seawater of mussel-kelp modes at different sampling time.

Source		df	Mean Squares	F	P
CO_2	Model	44	35.68	4.91	< 0.001
	Ratio	8	75.70	10.42	< 0.001
	Time	4	82.90	11.41	< 0.001
	Ratio* Time	32	19.78	2.72	< 0.001
	Error	90	7.27		
	Total	135			
CO_3^{2-}	Model	44	1120.02	52.75	< 0.001
	Ratio	8	2643.88	124.53	< 0.001
	Time	4	3040.75	143.22	< 0.001
	Ratio* Time	32	498.96	23.50	< 0.001
	Error	90	21.23		
	Total	135			
HCO_3^-	Model	44	11,471.26	11.05	< 0.001
	Ratio	8	13,956.99	13.44	< 0.001
	Time	4	40,096.91	38.62	< 0.001
	Ratio* Time	32	7271.63	7.00	< 0.001
	Error	90	1038.36		
	Total	135			
DIC	Model	44	9449.27	9.10	< 0.001
	Ratio	8	8558.58	8.24	< 0.001
	Time	4	37,894.38	36.49	< 0.001
	Ratio* Time	32	6116.30	5.89	< 0.001
	Error	90	1038.38		
	Total	135			

no significant difference between the initial and final pH in any treatment (one-way ANOVA: 6:3, $F_{4,10} = 3.32$, $P > 0.05$; 6:2, $F_{4,10} = 1.98$, $P > 0.05$; 6:1, $F_{4,10} = 1.10$, $P > 0.05$; 4:3, $F_{4,10} = 3.10$, $P > 0.05$; 4:2, $F_{4,10} = 1.03$, $P > 0.05$; 4:1, $F_{4,10} = 0.35$, $P > 0.05$; 2:3, $F_{4,10} = 0.80$, $P > 0.05$; 2:2, $F_{4,10} = 0.80$, $P > 0.05$; 2:1, $F_{4,10} = 0.44$, $P > 0.05$). The pH was generally higher in treatments with higher proportions of kelp, and decreased when there were more mussels. At the end of the experiment, the treatment with a mussel-kelp ratio of 2:3 exhibited the highest pH, while a ratio of 6:1 resulted in the lowest.

3.3. CO_2 system

The changes in dissolved inorganic carbon (DIC) and HCO_3^- followed similar trends (Fig. 3A, B). With the exception of the ratio 6:3 treatment, changes in DIC and HCO_3^- were generally small but significant. DIC and HCO_3^- had significantly decreased by the end of the experiment in treatments with mussel-kelp ratios of 2:1 and 6:3 (one-way ANOVA of DIC: 2:1, $F_{4,10} = 17.49$, $P < 0.05$; 6:3, $F_{4,10} = 28.49$, $P < 0.05$; one-way

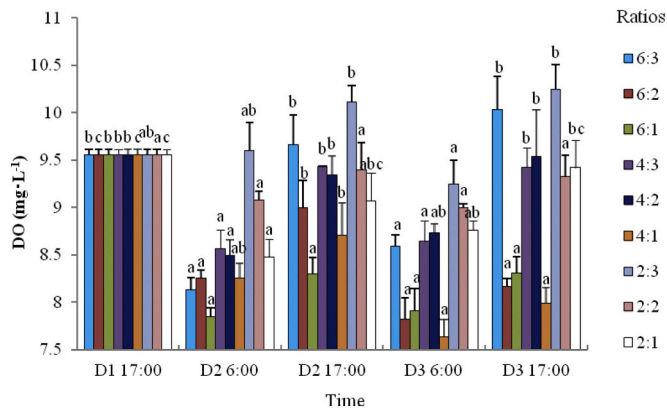


Fig. 1. The dissolved oxygen (DO) of different mussel-kelp proportions at different time points. D1 17:00, day one at 17:00 (initial); D2 6:00, day two at 6:00; D2 17:00, day two at 17:00; D3 6:00, day three at 6:00; D3 17:00, day three at 17:00. For details on the mussels kelp ratios, refer to Table 1.

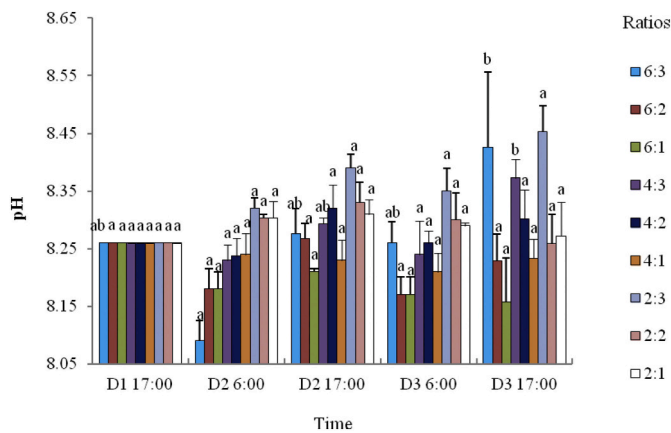


Fig. 2. The pH of different mussel-kelp ratios treatments at different time points. For the abbreviations, please refer to Fig. 1 and Table 1.

ANOVA of HCO_3^- : 2:1, $F_{4,10} = 15.23$, $P < 0.05$; 6:3, $F_{4,10} = 41.66$, $P < 0.05$). Significant increases in DIC concentrations were observed by the end of the experiment when those ratios were 6:1, 4:3, 4:1, and 2:3 (one-way ANOVA: 6:1, $F_{4,10} = 8.00$, $P < 0.05$; 4:3, $F_{4,10} = 2.73$, $P > 0.05$; 4:1, $F_{4,10} = 10.82$, $P < 0.05$; 2:3, $F_{4,10} = 3.12$, $P > 0.05$). The HCO_3^- concentration had significantly increased in treatments with the ratios 6:1, 4:1, and 2:2 at the end of the experiment (one-way ANOVA: 6:1, $F_{4,10} = 8.95$, $P < 0.05$; 4:1, $F_{4,10} = 10.38$, $P < 0.05$; 2:2, $F_{4,10} = 3.43$, $P > 0.05$). In the ratio 6:3 treatment, the concentrations of DIC and HCO_3^- peaked at $2470.38 \mu\text{mol L}^{-1}$ and $2341.33 \mu\text{mol L}^{-1}$ on day two at 6:00, respectively. Interestingly, the lowest observed DIC and HCO_3^- concentrations also occurred in the treatment with ratio 6:3 ($2177.40 \mu\text{mol L}^{-1}$ and $1981.34 \mu\text{mol L}^{-1}$, respectively), but at the end of the experiment.

Concentrations of CO_3^{2-} increased in all treatments, with the exception of treatments with ratios 6:1 and 4:1 which showed a significant decrease in CO_3^{2-} by the end of the experiment (one-way ANOVA: 6:1, $F_{4,10} = 8.30$, $P < 0.05$; 4:1, $F_{4,10} = 2.54$, $P > 0.05$). Generally, the level of CO_3^{2-} was higher when the proportion of kelp was higher and exhibited diurnal fluctuations (decreasing during the night and increasing during the day) (Fig. 3C).

The concentrations of CO_2 were higher in treatments with higher proportions of mussels. There was an observed increase over time when the mussel-kelp ratios were 4:1 and 6:1, yet for both ratios, there was no significant difference on day one at D1 17:00 and on day 3 at 17:00 (one-way ANOVA: 6:1, $F_{4,10} = 1.27$, $P > 0.05$; 4:1, $F_{4,10} = 1.42$, $P > 0.05$). By the end of the experiment, the concentration of CO_2 had significantly

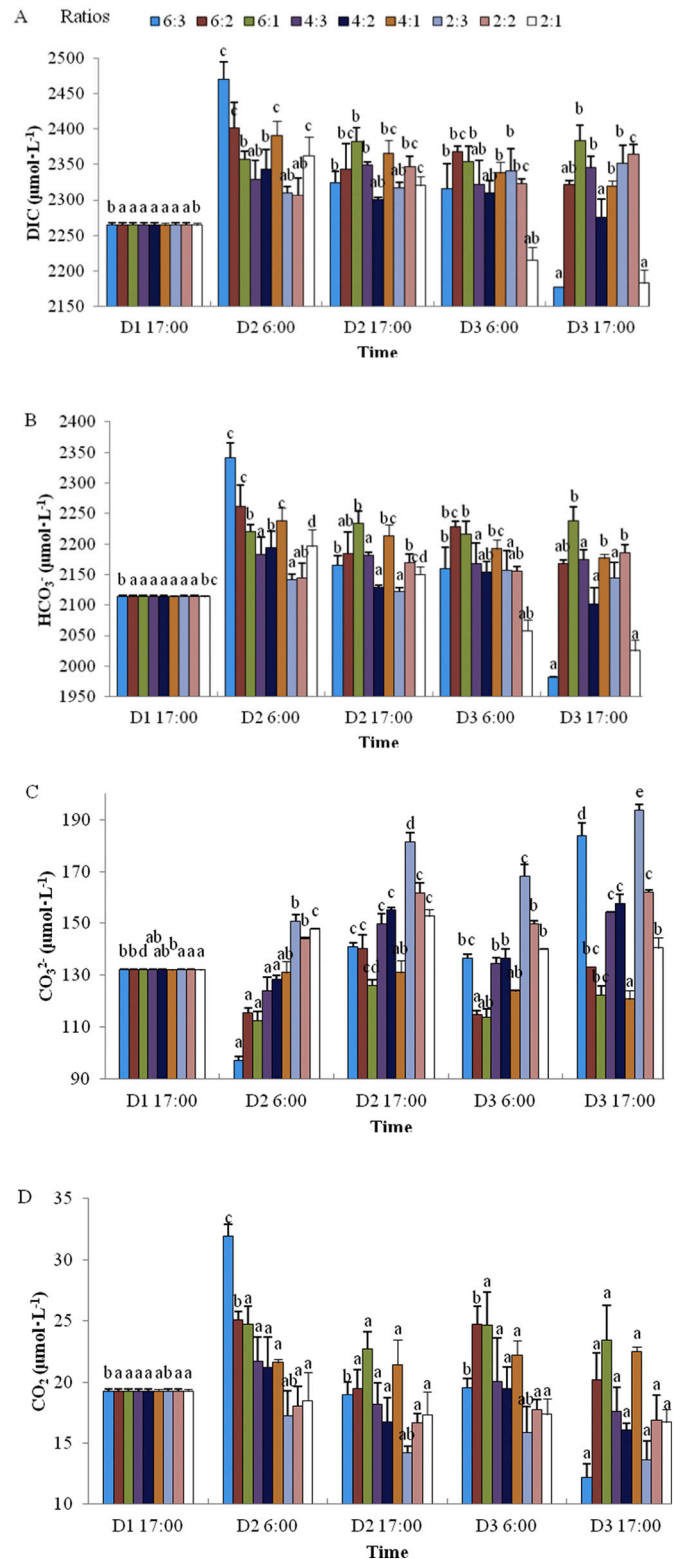


Fig. 3. The variations of parameters of the CO_2 system in different ratios of mussel-kelp (A) DIC (dissolved inorganic carbon), (B) HCO_3^- , (C) CO_3^{2-} , (D) CO_2 . For the abbreviations, please refer to Fig. 1 and Table 1.

decreased in treatments with ratios 2:3 and 6:3 (one-way ANOVA: 2:3, $F_{4,10} = 2.20$, $P > 0.05$; 6:3, $F_{4,10} = 65.34$, $P < 0.05$, Fig. 3D).

3.4. $p\text{CO}_2$

The changes in $p\text{CO}_2$ were very similar to those observed for CO_2 . Levels of $p\text{CO}_2$ also exhibited diurnal fluctuations, but in this case they increased during the night and decreasing during the day. Higher $p\text{CO}_2$ concentrations were observed in treatments with higher proportions of mussels, with significant increases observed when the ratios were 4:1 and 6:1 at the end of the experiment (one-way ANOVA: 6:1, $F_{4,10} = 8.18$, $P < 0.05$; 4:1, $F_{4,10} = 2.52$, $P > 0.05$, Fig. 4). The $p\text{CO}_2$ values in treatments with mussel-kelp ratios of 2:1 and 6:2 were similar at the beginning and the end of the experiment (one-way ANOVA: 2:1, $F_{4,10} = 1.61$, $P > 0.05$; 6:2, $F_{4,10} = 13.64$, $P < 0.05$), while the $p\text{CO}_2$ values in treatments with ratios 6:3, 4:3, 4:2, 2:3, and 2:2 had significantly decreased by the end of the experiment (one-way ANOVA: 6:3, $F_{4,10} = 52.14$, $P < 0.05$; 4:3, $F_{4,10} = 12.49$, $P < 0.05$; 4:2, $F_{4,10} = 50.51$, $P < 0.05$; 2:3, $F_{4,10} = 6.97$, $P < 0.05$; 2:2, $F_{4,10} = 2.73$, $P > 0.05$). Although $p\text{CO}_2$ reached a concentration of 694.90 μatm on day 2 at D2 6:00 when the mussel-kelp ratio was 6:3, it had decreased to 274.02 μatm by the end of the experiment, which was the lowest measured value for this parameter.

4. Discussion

Integrated multi-trophic aquaculture (IMTA) is an approach that can optimize management of an ecosystem, while considering site specificity, operational limits, and revenues alongside food safety guidelines and environmental quality and regulations. IMTA generally use combinations of fed aquaculture species (mainly finfish), organic particulate extractive aquaculture species (mainly suspension- and deposit-feeders), and/or inorganic extractive aquaculture species (mainly seaweed) (Troell et al., 2009). The IMTA shellfish-seaweed mode includes just two units, an organic particulate extractive aquaculture species and an inorganic extractive aquaculture species, this is the mode used in this study. All parameters measured in this study were significantly affected by the mussel-kelp ratio and sampling time ($P < 0.05$). DO oscillated between day and night but showed an overall increasing trend with increasing proportions of kelp. The diurnal cycle was caused by the mussels consuming oxygen during the night, and the kelp producing oxygen via photosynthesis during the day. Consequently, based on the observed changes in DO, it may appear that most appropriate proportions of mussel to kelp were suitable for aquaculture, with the exceptions being 4:1, 6:2 and 6:1. Similarly, fluctuations of pH, which generally correspond to changes in $p\text{CO}_2$ (Sabine et al., 2004), were weak and showed no significant decreases by the end of the experiment. However, the pH in treatments with mussel-kelp ratios of 4:1, 6:2 and 6:1 gradually decreased during the experiment, indicating a possible risk

of environmental acidification when the proportion of mussels is too high. At the same time, the $p\text{CO}_2$ concentration had significantly increased by the end of the experiment when the ratios of mussels to kelp were 4:1 and 6:1 ratios. Thus, considering the observed levels of pH and $p\text{CO}_2$, the 4:1, 6:2, and 6:1 mussel-kelp ratios appeared unsuitable for this IMTA mode. This was consistent with the results from the DO measurements.

Generally, it appeared that the greater the proportions of kelp in the mussel-kelp co-culture, the greater the environmental benefit. Indeed, the kelp absorbs CO_2 and nutrients, and produces oxygen, which can help in alleviating ocean acidification and eutrophication. However, the IMTA system is complex. Furthermore, the 'appropriate proportions' of different species within an IMTA system can change depending on the goals of that IMTA (Buschmann et al., 2001; Troell et al., 2003; Fang et al., 2019). For example, in order to achieve the highest nutrient uptake rate, the most suitable biomass proportion between a scallop (*C. farreri*) and a red alga (*G. lemaneiformis*) ranged from 1:0.33 to 1:0.80. However, according to Mao et al. (2009), only the 1:0.33 ratio also resulted in the highest specific growth rate (SGR). In this study, the concentration of DIC had increased significantly by the end of the experiment when mussel-kelp ratios were 6:1, 4:3, 4:1 and 2:3 ($P < 0.05$), while HCO_3^- concentrations increased significantly when the ratios were 6:1, 4:1 and 2:2 ($P < 0.05$). The differences in the changes of DIC and HCO_3^- from those of CO_2 and $p\text{CO}_2$ demonstrate the difficulty in defining what an appropriate ratio of species would be. Therefore, an optimal proportion of species can only be established when the IMTA system is given a categorical purpose.

Shellfish calcification transforms DIC to CaCO_3 , which integrates carbon into the shell in a stable manner. However, the calcification formula: $\text{Ca}^{2+} + 2\text{HCO}_3^- = \text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2$ shows that the amount of CO_2 released is equal to amount of carbon that is fixed in CaCO_3 . Frankignoulle et al. (1994) reported that only 0.56–0.81 mol of CO_2 was released for every 1.0 mol carbon that was fixed during calcification when the CO_2 concentration was 300–500 μatm . In addition to the CO_2 produced during calcification, shellfish respiration also releases CO_2 . In the shellfish-seaweed mode, the extra CO_2 released by shellfish is absorbed by seaweed, which makes the system a net carbon sink due to the carbon embedded in the shells. At the same time, the seaweed itself is able to fix carbon. However, to create the strongest net carbon sink without causing any negative effects on the environment, the appropriate ratios of shellfish to seaweed must be determined. In the present study, the $p\text{CO}_2$ had increased significantly by the end of the experiment when the ratios were skewed toward more mussels, such as when the ratios were 4:1 and 6:1 as the kelp was not able to take up all the CO_2 released by mussels. This resulted in a decrease in the DO and pH which can negatively affecting the IMTA environment. However, the $p\text{CO}_2$ had decreased significantly by the end of the experiment when the mussel-kelp ratios leaned toward more seaweed, such as when the ratios were 2:3, 4:3 and 6:3. This implied that the CO_2 released by the mussels was insufficient for the kelp absorption. Therefore, more mussels would be needed to obtain a more stable net carbon sink for the system. By the end of the experiment, based on the one-way ANOVA for $p\text{CO}_2$ and CO_2 in treatments with the different mussel-kelp, the mussel-kelp ratios best suited for maintaining system health as well as a net carbon sink were 2:1, 2:2, 4:2 and 4:3.

5. Conclusions

The shellfish-seaweed IMTA mode is an advantageous system that uses only extractive aquaculture species that have the ability to remove nutrients and carbon from sea water. However, optimal proportions of shellfish and seaweed vary depending on the goals of the system. In the present study, the optimal ratios of mussel to kelp (2:1, 2:2, 2:3, 4:2, 4:3, and 6:3) ratios were determined according to measured DO, pH and $p\text{CO}_2$ fluctuations, choosing those that were beneficial to the environment of Sanggou Bay. Generally, it appeared that higher proportions of

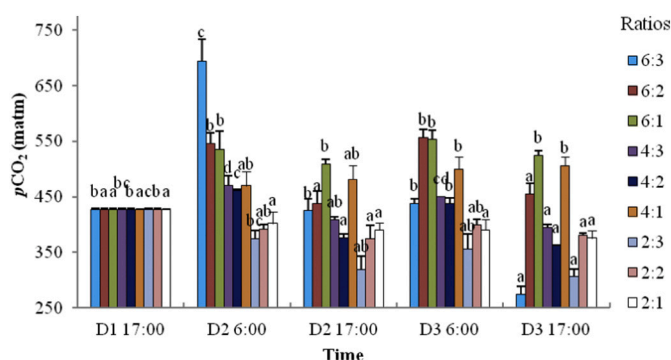


Fig. 4. The fluctuations of $p\text{CO}_2$ in different ratios of mussel-kelp at different time points. For the abbreviations, please refer to Fig. 1 and Table 1.

kelp in the mussel-kelp mode had a positive effect on the environmental parameters. However, if the effects on the CO₂ system are prioritized, and the goal of the system is to create a net carbon sink through calcification, the range of suitable ratios is narrowed. Therefore, in this study, with the goal of creating the most effective net carbon sink possible while reduction any negative effects on the environment, the appropriate mussel-kelp ratios were 2:1, 2:2, 4:2 and 4:3.

Declaration of Competing Interest

We declared that we have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted which is entitled “Effects of mussel-kelp ratios in integrated mariculture on the carbon dioxide system in Sanggou Bay”.

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